



ENVIRONMENTAL SCIENCE

COMPLETE COURSEBOOK

Ecology - Biodiversity - Resources - Pollution - Climate

Energy - Conservation - Sustainability - Environmental Policy

- Clear explanations in English
- Scientific concepts with global and Madagascar examples
- Tables, case studies, exercises and exam practice
- Revision tests, answer guidance and quick-reference pages

PURPOSE

This coursebook introduces environmental science as an interdisciplinary subject linking ecology, Earth systems, biodiversity, natural resources, pollution, climate, energy, sustainability, policy and environmental research. Explanations are in English and examples are drawn from many regions, including Madagascar.

HOW TO STUDY

- Begin with the guiding idea on each page.
- Learn the key terms and use them in complete explanations.
- Trace cause-and-effect chains instead of memorizing isolated facts.
- Compare environmental options using evidence, benefits, costs and trade-offs.
- Practice interpreting graphs, maps, tables and field observations.

ASSESSMENT SKILLS

- Define accurately.
- Explain mechanisms step by step.
- Analyze environmental data.
- Evaluate options and trade-offs.
- Apply concepts to real case studies.
- Design simple investigations and monitoring plans.

FOUNDATIONS - PAGES 5-14

- What Is Environmental Science?
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What Is Environmental Science?

Complete Environmental Science Coursebook - explanations in English

FOUNDATIONS

GUIDING IDEA

Environmental science studies interactions among natural systems, human societies and technology in order to understand environmental problems and possible responses.

CORE CONCEPTS

- It is interdisciplinary: ecology, chemistry, geology, geography, economics, sociology and policy all contribute.
- Environmental questions often involve trade-offs among ecological protection, human well-being and economic activity.
- Evidence can come from field observations, experiments, long-term monitoring, models and historical records.
- A systems approach asks how parts interact rather than studying each part in isolation.

Key term	Meaning	Context
environment	surroundings and conditions affecting organisms and people	A river basin links rainfall, soils, farms, cities, wetlands and
interdisciplinary	combining knowledge from several fields	Urban air quality combines atmospheric chemistry, transport
system	connected parts that interact	Urban air quality combines atmospheric chemistry, transport
evidence	observations or data used to support a conclusion	Urban air quality combines atmospheric chemistry, transport

REAL-WORLD EXAMPLES

- A river basin links rainfall, soils, farms, cities, wetlands and downstream fisheries.
- Urban air quality combines atmospheric chemistry, transport patterns, health data and policy.

FIELD / DATA TASK

Choose one local environmental issue and list the scientific, social and economic information needed to study it.

State the variable or evidence you would collect.

Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of What Is Environmental Science?.

Define two key terms from this page.

Give one environmental benefit or risk linked to this topic.

Apply the concept to a local or Madagascar example.

GUIDING IDEA

Earth can be understood through interacting spheres: atmosphere, hydrosphere, lithosphere and biosphere.

CORE CONCEPTS

- Energy from the Sun drives climate, photosynthesis and many surface processes.
- Matter cycles between reservoirs; it is transformed and moved rather than simply disappearing.
- Feedback loops can amplify change or stabilize a system.
- Human activity has become an important force altering flows of carbon, nitrogen, water and materials.

Key term	Meaning	Context
atmosphere	layer of gases surrounding Earth	Deforestation can change carbon storage, local water cycling
hydrosphere	Earth's water in all forms	A drought can affect vegetation, fire risk, reservoirs and food
lithosphere	solid outer part of Earth	A drought can affect vegetation, fire risk, reservoirs and food
biosphere	all regions where life exists	A drought can affect vegetation, fire risk, reservoirs and food

REAL-WORLD EXAMPLES

- Deforestation can change carbon storage, local water cycling and soil erosion at the same time.
- A drought can affect vegetation, fire risk, reservoirs and food production.

FIELD / DATA TASK

Draw a four-sphere diagram and add three arrows showing interactions among the spheres.

State the variable or evidence you would collect.

Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Earth as a System.

Define two key terms from this page.

Give one environmental benefit or risk linked to this topic.

Apply the concept to a local or Madagascar example.

GUIDING IDEA

Ecologists study organisms, populations, communities, ecosystems, biomes and the biosphere at different scales.

CORE CONCEPTS

- A population contains members of one species in an area.
- A community includes populations of different species that interact.
- An ecosystem includes living organisms and the nonliving environment.
- Scale matters because a pattern visible at one level may be hidden at another.

Key term	Meaning	Context
organism	individual living thing	A lemur population lives within a forest community that also
population	members of one species in an area	A coral reef ecosystem includes organisms plus water
community	interacting populations of different species	A coral reef ecosystem includes organisms plus water
ecosystem	organisms plus their physical environment	A coral reef ecosystem includes organisms plus water

REAL-WORLD EXAMPLES

- A lemur population lives within a forest community that also includes trees, insects, birds and fungi.
- A coral reef ecosystem includes organisms plus water chemistry, light and substrate.

FIELD / DATA TASK

Classify five examples from your area into organism, population, community, ecosystem or biome.

State the variable or evidence you would collect.

Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Ecological Levels of Organization.

Define two key terms from this page.

Give one environmental benefit or risk linked to this topic.

Apply the concept to a local or Madagascar example.

GUIDING IDEA

Energy enters most ecosystems as sunlight and passes through trophic levels while much is lost as heat.

CORE CONCEPTS

- Producers capture energy, consumers obtain it by feeding, and decomposers recycle nutrients.
- Food chains simplify feeding relations; food webs show multiple connections.
- Ecological pyramids can represent numbers, biomass or energy.
- Because energy transfer is inefficient, higher trophic levels generally support less biomass.

Key term	Meaning	Context
producer	organism that makes organic matter from external energy	Grass -> grasshopper -> frog -> snake is a simple food chain.
consumer	organism obtaining energy by feeding on others	A forest food web contains many overlapping predator-prey
decomposer	organism breaking down dead material	A forest food web contains many overlapping predator-prey
trophic level	feeding position in a food chain or web	A forest food web contains many overlapping predator-prey

REAL-WORLD EXAMPLES

- Grass -> grasshopper -> frog -> snake is a simple food chain.
- A forest food web contains many overlapping predator-prey links.

FIELD / DATA TASK

Construct a food web with at least six organisms and label each trophic role.

State the variable or evidence you would collect.

Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Energy Flow in Ecosystems.

Define two key terms from this page.

Give one environmental benefit or risk linked to this topic.

Apply the concept to a local or Madagascar example.

GUIDING IDEA

Water, carbon, nitrogen and phosphorus move through organisms, air, water, rocks and soils.

CORE CONCEPTS

- The carbon cycle links photosynthesis, respiration, decomposition, oceans and combustion.
- The nitrogen cycle depends strongly on microorganisms that fix, transform and return nitrogen compounds.
- Phosphorus has no major atmospheric phase and often moves slowly through rocks, soils and water.
- Human activities can accelerate transfers and create imbalances such as eutrophication.

Key term	Meaning	Context
carbon cycle	movement of carbon among major reservoirs	Fertilizer runoff can add nitrogen and phosphorus to lakes.
nitrogen cycle	movement and transformation of nitrogen	Burning fossil fuels moves carbon from geological storage into
phosphorus cycle	movement of phosphorus through rocks, soil, water and	Burning fossil fuels moves carbon from geological storage into
reservoir	stored body of water behind a barrier	Burning fossil fuels moves carbon from geological storage into

REAL-WORLD EXAMPLES

- Fertilizer runoff can add nitrogen and phosphorus to lakes.
- Burning fossil fuels moves carbon from geological storage into the atmosphere.

FIELD / DATA TASK

Make a two-column list of natural processes and human processes that alter the carbon cycle.

State the variable or evidence you would collect.

Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Biogeochemical Cycles.

Define two key terms from this page.

Give one environmental benefit or risk linked to this topic.

Apply the concept to a local or Madagascar example.

GUIDING IDEA

Population size changes through births, deaths, immigration and emigration, and is limited by resources and environmental conditions.

CORE CONCEPTS

- Exponential growth is possible when resources are abundant for a period.
- Logistic growth includes a carrying capacity that reflects limiting conditions.
- Density-dependent factors often become stronger as populations become crowded.
- Population models simplify reality but help explain trends and test scenarios.

Key term	Meaning	Context
population growth	change in population size over time	A deer population may grow rapidly after predators decline, then
carrying capacity	level of use a site can sustain without unacceptable impact	An invasive plant may spread quickly when it first enters suitable
limiting factor	resource or condition restricting growth	An invasive plant may spread quickly when it first enters suitable
density dependence	effect that changes with population density	An invasive plant may spread quickly when it first enters suitable

REAL-WORLD EXAMPLES

- A deer population may grow rapidly after predators decline, then slow as food becomes limiting.
- An invasive plant may spread quickly when it first enters suitable habitat.

FIELD / DATA TASK

Sketch exponential and logistic growth curves and explain why they differ.

State the variable or evidence you would collect.

Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Population Ecology.

Define two key terms from this page.

Give one environmental benefit or risk linked to this topic.

Apply the concept to a local or Madagascar example.

GUIDING IDEA

Competition, predation, herbivory, parasitism, mutualism and commensalism shape ecological communities.

CORE CONCEPTS

- Competition occurs when organisms use a resource that is insufficient for all.
- Predators can influence prey numbers and behavior, sometimes affecting whole food webs.
- Mutualistic interactions benefit both partners and can be essential for pollination or nutrient exchange.
- Species interactions change with environmental context and are not always fixed.

Key term	Meaning	Context
competition	use of a limited resource by multiple organisms	Pollinators and flowering plants often have mutualistic
predation	one organism kills and consumes another	Ticks benefit from feeding on mammals while harming the host.
mutualism	interaction benefiting both partners	Ticks benefit from feeding on mammals while harming the host.
parasitism	one partner benefits while harming the host	Ticks benefit from feeding on mammals while harming the host.

REAL-WORLD EXAMPLES

- Pollinators and flowering plants often have mutualistic relationships.
- Ticks benefit from feeding on mammals while harming the host.

FIELD / DATA TASK

Give one local or familiar example for each of four interaction types.
State the variable or evidence you would collect.
Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Species Interactions.
Define two key terms from this page.
Give one environmental benefit or risk linked to this topic.
Apply the concept to a local or Madagascar example.

GUIDING IDEA

Communities change through time after new surfaces form or disturbances alter existing ecosystems.

CORE CONCEPTS

- Primary succession begins where soil is absent or extremely limited.
- Secondary succession occurs after disturbance where soil and biological legacies remain.
- Succession does not always lead to one permanent final state; repeated disturbance can maintain mosaics.
- Management decisions may aim to allow, redirect or periodically reset succession.

Key term	Meaning	Context
succession	change in community composition through time	Vegetation recovery after a fire can be secondary succession.
primary succession	succession beginning with little or no soil	Bare volcanic rock may gradually develop soil and vegetation.
secondary succession	succession after disturbance where soil remains	Bare volcanic rock may gradually develop soil and vegetation.
disturbance	disruption of normal activity or ecological process	Bare volcanic rock may gradually develop soil and vegetation.

REAL-WORLD EXAMPLES

- Vegetation recovery after a fire can be secondary succession.
- Bare volcanic rock may gradually develop soil and vegetation.

FIELD / DATA TASK

Describe how vegetation might change during 30 years after farmland is abandoned.

State the variable or evidence you would collect.

Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Ecological Succession.

Define two key terms from this page.

Give one environmental benefit or risk linked to this topic.

Apply the concept to a local or Madagascar example.

GUIDING IDEA

A niche describes the environmental conditions and resource use that allow a species to persist.

CORE CONCEPTS

- Adaptations are inherited traits that improve survival or reproduction in particular environments.
- The fundamental niche can be broader than the realized niche because competition and predators restrict actual use.
- Specialist species often depend on narrow conditions, while generalists tolerate wider ranges.
- Rapid environmental change may outpace the ability of populations to adapt.

Key term	Meaning	Context
niche	conditions and resource use allowing a species to persist	Cacti show adaptations for water conservation.
adaptation	adjustment reducing harm from climate impacts	Many island species evolve narrow niches and can be
specialist	species with narrow ecological requirements	Many island species evolve narrow niches and can be
generalist	species with broad ecological tolerance	Many island species evolve narrow niches and can be

REAL-WORLD EXAMPLES

- Cacti show adaptations for water conservation.
- Many island specialist evolve narrow niches and can be vulnerable to introduced competitors.

FIELD / DATA TASK

Compare one specialist and one generalist species and explain the advantages and risks of each strategy.
State the variable or evidence you would collect.
Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Ecological Niches and Adaptation.
Define two key terms from this page.
Give one environmental benefit or risk linked to this topic.
Apply the concept to a local or Madagascar example.

GUIDING IDEA

Ecosystems provide benefits such as food, clean water, climate regulation, soil formation, pollination and cultural value.

CORE CONCEPTS

- Provisioning services include products such as timber, fish and freshwater.
- Regulating services include flood control, carbon storage and water purification.
- Supporting processes such as nutrient cycling make many other services possible.
- Valuation can inform decisions but not all ecological or cultural values are easily expressed in money.

Key term	Meaning	Context
ecosystem service	benefit people obtain from ecosystems	Mangroves can support fisheries, reduce coastal erosion and
provisioning	service supplying material goods	Urban trees can provide shade, habitat and stormwater benefits.
regulating	service moderating environmental conditions	Urban trees can provide shade, habitat and stormwater benefits.
cultural value	nonmaterial benefit from nature	Urban trees can provide shade, habitat and stormwater benefits.

REAL-WORLD EXAMPLES

- Mangroves can support fisheries, reduce coastal erosion and store carbon.
- Urban trees can provide shade, habitat and stormwater benefits.

FIELD / DATA TASK

Identify five ecosystem services supplied by one forest, wetland, coast or urban green space.

State the variable or evidence you would collect.

Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Ecosystem Services.

Define two key terms from this page.

Give one environmental benefit or risk linked to this topic.

Apply the concept to a local or Madagascar example.

GUIDING IDEA

Biodiversity includes genetic diversity, species diversity and ecosystem diversity.

CORE CONCEPTS

- Genetic diversity can help populations adapt to disease and environmental change.
- Species richness counts species, while evenness considers their relative abundance.
- Ecosystem diversity reflects variation in habitats and ecological processes.
- Biodiversity has ecological, economic, cultural, scientific and intrinsic values.

Key term	Meaning	Context
biodiversity	variety of genes, species and ecosystems	Madagascar contains many endemic species found nowhere
genetic diversity	variation in genes within a species	A farm growing one crop variety has less genetic diversity than a
species richness	number of species present	A farm growing one crop variety has less genetic diversity than a
evenness	similarity in relative abundance among species	A farm growing one crop variety has less genetic diversity than a

REAL-WORLD EXAMPLES

- Madagascar contains many endemic species found nowhere else.
- A farm growing one crop variety has less genetic diversity than a system with many local varieties.

FIELD / DATA TASK

Explain why conserving genetic diversity can be important even when a species is not currently rare.

State the variable or evidence you would collect.

Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Understanding Biodiversity.

Define two key terms from this page.

Give one environmental benefit or risk linked to this topic.

Apply the concept to a local or Madagascar example.

GUIDING IDEA

Endemic species occur naturally in restricted geographic areas, making some regions especially important for conservation.

CORE CONCEPTS

- Islands often have high endemism because isolation promotes evolutionary divergence.
- A biodiversity hotspot combines exceptional biological richness with high levels of habitat loss.
- Hotspot status helps prioritize attention but does not mean other ecosystems are unimportant.
- Conservation planning should consider species, habitats, connectivity and local communities.

Key term	Meaning	Context
endemic	naturally restricted to a particular area	Madagascar is globally known for high endemism in lemurs,
hotspot	region with high biodiversity and severe habitat loss	Mountain ranges can create isolated habitats with locally
island biogeography	study of species patterns on islands and isolated habitats	Mountain ranges can create isolated habitats with locally
range	geographic area occupied by a species	Mountain ranges can create isolated habitats with locally

REAL-WORLD EXAMPLES

- Madagascar is globally known for high endemism in lemurs, reptiles and plants.
- Mountain ranges can create isolated habitats with locally endemic species.

FIELD / DATA TASK

Why can a species with a small geographic range be more vulnerable than a widespread species?

State the variable or evidence you would collect.

Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Endemism and Hotspots.

Define two key terms from this page.

Give one environmental benefit or risk linked to this topic.

Apply the concept to a local or Madagascar example.

GUIDING IDEA

Habitat loss, overexploitation, invasive species, pollution and climate change are major drivers of biodiversity decline.

CORE CONCEPTS

- Habitat fragmentation divides continuous ecosystems into smaller patches.
- Overharvesting can reduce populations faster than reproduction replaces individuals.
- Invasive species may compete with, prey on or spread disease to native species.
- Multiple threats can interact and make conservation more difficult.

Key term	Meaning	Context
habitat loss	destruction or conversion of habitat	Roads can fragment forests and increase human access.
fragmentation	division of habitat into smaller isolated patches	Introduced rats and cats have harmed many island bird
overexploitation	use faster than natural replacement	Introduced rats and cats have harmed many island bird
invasive species	non-native species causing ecological or economic harm	Introduced rats and cats have harmed many island bird

REAL-WORLD EXAMPLES

- Roads can fragment forests and increase human access.
- Introduced rats and cats have harmed many island bird populations.

FIELD / DATA TASK

Create a threat matrix for one species, showing direct threats and indirect drivers.

State the variable or evidence you would collect.

Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Threats to Biodiversity.

Define two key terms from this page.

Give one environmental benefit or risk linked to this topic.

Apply the concept to a local or Madagascar example.

GUIDING IDEA

Protected areas aim to conserve ecosystems, species and ecological processes while allowing different levels of human use.

CORE CONCEPTS

- Effective protection requires clear goals, management capacity and long-term financing.
- Ecological corridors can connect isolated habitats and support movement and gene flow.
- Local communities can be crucial partners in conservation and resource management.
- Protection on paper alone may be insufficient without enforcement and social legitimacy.

Key term	Meaning	Context
protected area	geographic area managed for conservation	A national park may protect core habitat while a surrounding
corridor	habitat link connecting larger areas	Wildlife corridors can link forest fragments.
buffer zone	area surrounding a core zone with compatible uses	Wildlife corridors can link forest fragments.
management	planned actions to achieve goals	Wildlife corridors can link forest fragments.

REAL-WORLD EXAMPLES

- A national park may protect core habitat while a surrounding buffer zone supports compatible livelihoods.
- Wildlife corridors can link forest fragments.

FIELD / DATA TASK

Design a simple protected-area zoning plan with core, buffer and community-use areas.
State the variable or evidence you would collect.
Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Protected Areas.
Define two key terms from this page.
Give one environmental benefit or risk linked to this topic.
Apply the concept to a local or Madagascar example.

GUIDING IDEA

In situ conservation protects species in natural habitats, while ex situ conservation maintains them outside the wild.

CORE CONCEPTS

- In situ methods conserve ecological interactions as well as target species.
- Zoos, botanical gardens, seed banks and captive breeding are examples of ex situ methods.
- Ex situ conservation can provide insurance but cannot replace functioning ecosystems.
- Reintroduction requires suitable habitat, genetic planning and control of original threats.

Key term	Meaning	Context
in situ	conservation in natural habitat	Seed banks preserve genetic material from crop relatives and
ex situ	conservation outside natural habitat	Captive breeding can support recovery when wild populations
seed bank	facility storing seeds for future conservation or use	Captive breeding can support recovery when wild populations
captive breeding	managed reproduction outside the wild	Captive breeding can support recovery when wild populations

REAL-WORLD EXAMPLES

- Seed banks preserve genetic material from crop relatives and threatened plants.
- Captive breeding can support recovery when wild populations are critically small.

FIELD / DATA TASK

Compare two strengths and two limitations of in situ and ex situ conservation.

State the variable or evidence you would collect.

Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of In Situ and Ex Situ Conservation.

Define two key terms from this page.

Give one environmental benefit or risk linked to this topic.

Apply the concept to a local or Madagascar example.

GUIDING IDEA

Conservation can create conflict when wildlife damages crops, livestock or property, or when protection restricts access to resources.

CORE CONCEPTS

- Human-wildlife conflict has ecological and social dimensions.
- Prevention can include barriers, guarding, compensation, insurance or land-use planning.
- Local knowledge can improve practical solutions.
- Fair distribution of conservation costs and benefits can strengthen long-term support.

Key term	Meaning	Context
human-wildlife conflict	negative interaction between people and wildlife	Elephants may damage crops near protected areas.
compensation	payment or support for losses	Predators can threaten livestock where grazing overlaps with
coexistence	people and wildlife sharing landscapes with manageable	Predators can threaten livestock where grazing overlaps with
livelihood	means of obtaining food, income and necessities	Predators can threaten livestock where grazing overlaps with

REAL-WORLD EXAMPLES

- Elephants may damage crops near protected areas.
- Predators can threaten livestock where grazing overlaps with wildlife habitat.

FIELD / DATA TASK

Propose three ways to reduce conflict while protecting both wildlife and local livelihoods.
State the variable or evidence you would collect.
Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Wildlife Conservation and Human Conflict.
Define two key terms from this page.
Give one environmental benefit or risk linked to this topic.
Apply the concept to a local or Madagascar example.

GUIDING IDEA

Lemurs illustrate the importance of endemism, forest conservation and community-based approaches in Madagascar.

CORE CONCEPTS

- Lemurs occupy diverse ecological roles, including seed dispersal and herbivory.
- Many species depend on forest habitats that are fragmented or degraded.
- Conservation approaches may combine protected areas, restoration, research and livelihood support.
- Environmental education and locally designed management can complement formal protection.

Key term	Meaning	Context
lemur	endemic primate group of Madagascar	Different lemur species use rainforests, dry forests and spiny
endemic species	species naturally limited to a region	Forest restoration can reconnect habitat patches over time.
forest fragmentation	breaking continuous forest into smaller patches	Forest restoration can reconnect habitat patches over time.
restoration	assisting recovery of degraded ecosystems	Forest restoration can reconnect habitat patches over time.

REAL-WORLD EXAMPLES

- Different lemur species use rainforests, dry forests and spiny habitats.
- Forest restoration can reconnect habitat patches over time.

FIELD / DATA TASK

Explain why protecting a lemur species usually requires protecting more than the animals themselves.
State the variable or evidence you would collect.
Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Lemurs and Madagascar Conservation.
Define two key terms from this page.
Give one environmental benefit or risk linked to this topic.
Apply the concept to a local or Madagascar example.

GUIDING IDEA

Restoration seeks to recover ecological function, biodiversity or services in degraded ecosystems.

CORE CONCEPTS

- Passive restoration allows natural regeneration where disturbance pressure has been reduced.
- Active restoration may involve planting native species, erosion control or invasive-species removal.
- Reference ecosystems help define goals, but restoration rarely recreates the past exactly.
- Monitoring should track survival, diversity, soil condition and ecological function.

Key term	Meaning	Context
restoration	assisting recovery of degraded ecosystems	Assisted natural regeneration can be cheaper than planting
natural regeneration	recovery through spontaneous ecological processes	Riparian planting can stabilize riverbanks and improve habitat.
native species	species occurring naturally in an area	Riparian planting can stabilize riverbanks and improve habitat.
reference ecosystem	model used to guide restoration goals	Riparian planting can stabilize riverbanks and improve habitat.

REAL-WORLD EXAMPLES

- Assisted natural regeneration can be cheaper than planting every tree.
- Riparian planting can stabilize riverbanks and improve habitat.

FIELD / DATA TASK

Write four measurable indicators for a five-year forest restoration project.
State the variable or evidence you would collect.
Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Forest Restoration.
Define two key terms from this page.
Give one environmental benefit or risk linked to this topic.
Apply the concept to a local or Madagascar example.

GUIDING IDEA

Coral reefs, mangroves, seagrasses and coastal wetlands support biodiversity, fisheries and shoreline protection.

CORE CONCEPTS

- Overfishing can change food webs and reduce resilience.
- Mangrove clearing removes nursery habitat and natural coastal protection.
- Marine protected areas can support recovery when well designed and enforced.
- Land-based pollution can damage marine ecosystems far from its source.

Key term	Meaning	Context
coral reef	marine ecosystem built largely by corals	Sediment from eroded land can reduce light reaching coral
mangrove	salt-tolerant coastal forest	Mangroves provide nursery habitat for many fish species.
seagrass	marine flowering plant forming underwater meadows	Mangroves provide nursery habitat for many fish species.
marine protected area	marine zone managed for conservation	Mangroves provide nursery habitat for many fish species.

REAL-WORLD EXAMPLES

- Sediment from eroded land can reduce light reaching coral reefs.
- Mangroves provide nursery habitat for many fish species.

FIELD / DATA TASK

Trace one land-based activity through a chain of effects on a coastal ecosystem.
State the variable or evidence you would collect.
Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Marine and Coastal Conservation.
Define two key terms from this page.
Give one environmental benefit or risk linked to this topic.
Apply the concept to a local or Madagascar example.

GUIDING IDEA

Conservation decisions require scientific evidence but also involve values about species, ecosystems, people and future generations.

CORE CONCEPTS

- Prioritization may consider extinction risk, ecological importance, uniqueness, feasibility and cost.
- Flagship species can attract public support but may not represent all biodiversity needs.
- Environmental justice asks who receives benefits and who bears conservation costs.
- Transparent criteria make difficult choices easier to evaluate.

Key term	Meaning	Context
priority setting	choosing actions using explicit criteria	A rare species may need expensive protection while a larger
flagship species	charismatic species used to build support	Community-managed forests may combine conservation and
environmental ethics	study of moral relations involving nature	Community-managed forests may combine conservation and
justice	fair distribution of rights, benefits and burdens	Community-managed forests may combine conservation and

REAL-WORLD EXAMPLES

- A rare species may need expensive protection while a larger habitat could protect many common species.
- Community-managed forests may combine conservation and resource use.

FIELD / DATA TASK

Create five criteria for choosing between two proposed conservation projects.
State the variable or evidence you would collect.
Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Conservation Priorities and Ethics.
Define two key terms from this page.
Give one environmental benefit or risk linked to this topic.
Apply the concept to a local or Madagascar example.

GUIDING IDEA

Resources are classified partly by the rate at which natural processes replace them relative to human use.

CORE CONCEPTS

- Renewable does not mean unlimited; forests, fisheries and groundwater can be depleted if use exceeds renewal.
- Nonrenewable resources such as fossil fuels and metal ores form over geological timescales.
- Resource management compares stocks, flows, extraction rates and demand.
- Efficiency and circular use can reduce pressure on both resource types.

Key term	Meaning	Context
renewable resource	resource replenished on useful human timescales	A forest can be renewable under sustainable harvesting but
nonrenewable resource	resource replaced too slowly for human timescales	Oil is nonrenewable on human timescales.
stock	harvested population or group	Oil is nonrenewable on human timescales.
flow	rate of movement or use	Oil is nonrenewable on human timescales.

REAL-WORLD EXAMPLES

- A forest can be renewable under sustainable harvesting but degraded under excessive cutting.
- Oil is nonrenewable on human timescales.

FIELD / DATA TASK

Give two renewable resources that can still be depleted and explain why.

State the variable or evidence you would collect.

Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Renewable and Nonrenewable Resources.

Define two key terms from this page.

Give one environmental benefit or risk linked to this topic.

Apply the concept to a local or Madagascar example.

GUIDING IDEA

Freshwater availability depends on precipitation, storage, river flow, groundwater, infrastructure and demand.

CORE CONCEPTS

- Only a small fraction of Earth's water is easily accessible freshwater.
- Watersheds connect upstream land use with downstream water quantity and quality.
- Groundwater can provide reliable supplies but may recharge slowly.
- Integrated water management balances households, agriculture, ecosystems and industry.

Key term	Meaning	Context
watershed	land area draining to a common outlet	Deforestation upstream can affect sediment loads in rivers.
aquifer	permeable underground layer storing groundwater	Overpumping an aquifer can lower the water table.
groundwater	water stored below the land surface	Overpumping an aquifer can lower the water table.
recharge	addition of water to an aquifer	Overpumping an aquifer can lower the water table.

REAL-WORLD EXAMPLES

- Deforestation upstream can affect sediment loads in rivers.
- Overpumping an aquifer can lower the water table.

FIELD / DATA TASK

Draw a watershed and mark three possible sources of water pollution.
State the variable or evidence you would collect.
Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Freshwater Resources.
Define two key terms from this page.
Give one environmental benefit or risk linked to this topic.
Apply the concept to a local or Madagascar example.

GUIDING IDEA

Water quality is assessed using physical, chemical and biological indicators.

CORE CONCEPTS

- Turbidity, temperature, pH, dissolved oxygen and nutrient concentrations are common measurements.
- Pathogens from untreated waste can create serious health risks.
- High nutrient levels can cause algal blooms and oxygen depletion.
- Source protection is often cheaper than treating heavily polluted water later.

Key term	Meaning	Context
turbidity	cloudiness caused by suspended particles	Fertilizer runoff can raise nitrate and phosphate concentrations.
pH	measure of acidity or alkalinity	Warm polluted water may hold less dissolved oxygen than
dissolved oxygen	oxygen gas dissolved in water	Warm polluted water may hold less dissolved oxygen than
pathogen	disease-causing organism or agent	Warm polluted water may hold less dissolved oxygen than

REAL-WORLD EXAMPLES

- Fertilizer runoff can raise nitrate and phosphate concentrations.
- Warm polluted water may hold less dissolved oxygen than cooler water.

FIELD / DATA TASK

Explain how high nutrient input can eventually reduce oxygen available to fish.

State the variable or evidence you would collect.

Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Water Quality.

Define two key terms from this page.

Give one environmental benefit or risk linked to this topic.

Apply the concept to a local or Madagascar example.

GUIDING IDEA

Soil develops through interactions among parent material, climate, organisms, topography and time.

CORE CONCEPTS

- Soil texture depends on the proportions of sand, silt and clay.
- Organic matter improves nutrient storage, structure and water-holding capacity.
- Soil horizons record processes such as leaching, accumulation and biological activity.
- Healthy soil supports agriculture, biodiversity, water regulation and carbon storage.

Key term	Meaning	Context
soil horizon	distinct layer in a soil profile	Clay-rich soils retain water differently from sandy soils.
texture	relative amount of sand, silt and clay	Leaf litter contributes organic material to forest soils.
organic matter	carbon-rich material from living or once-living organisms	Leaf litter contributes organic material to forest soils.
leaching	downward removal of dissolved materials by water	Leaf litter contributes organic material to forest soils.

REAL-WORLD EXAMPLES

- Clay-rich soils retain water differently from sandy soils.
- Leaf litter contributes organic material to forest soils.

FIELD / DATA TASK

Compare how sandy and clayey soils differ in drainage and water retention.

State the variable or evidence you would collect.

Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Soil Formation and Properties.

Define two key terms from this page.

Give one environmental benefit or risk linked to this topic.

Apply the concept to a local or Madagascar example.

GUIDING IDEA

Erosion removes topsoil by water or wind and can reduce productivity, water quality and ecosystem function.

CORE CONCEPTS

- Vegetation cover protects soil from raindrop impact and slows runoff.
- Steep slopes, intense rainfall and bare ground increase erosion risk.
- Terracing, contour farming, mulching and agroforestry can reduce soil loss.
- Land degradation can include nutrient depletion, compaction, salinization and loss of vegetation.

Key term	Meaning	Context
erosion	removal and transport of soil or rock	Cultivating steep bare slopes can accelerate erosion.
topsoil	upper fertile soil layer	Riparian vegetation can trap sediment before it enters rivers.
runoff	water flowing over land surface	Riparian vegetation can trap sediment before it enters rivers.
land degradation	decline in land quality or productive capacity	Riparian vegetation can trap sediment before it enters rivers.

REAL-WORLD EXAMPLES

- Cultivating steep bare slopes can accelerate erosion.
- Riparian vegetation can trap sediment before it enters rivers.

FIELD / DATA TASK

Design a soil-conservation plan for a sloping farm exposed to heavy rainfall.
State the variable or evidence you would collect.
Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Soil Erosion and Land Degradation.
Define two key terms from this page.
Give one environmental benefit or risk linked to this topic.
Apply the concept to a local or Madagascar example.

GUIDING IDEA

Forests provide timber, fuel, food, habitat, water regulation, carbon storage and cultural services.

CORE CONCEPTS

- Tropical, temperate and boreal forests differ in climate, species and productivity.
- Deforestation converts forest to another land use, while degradation reduces forest quality without complete removal.
- Sustainable forestry can include reduced-impact harvesting, regeneration and protected areas.
- Forest management must consider ecology, local livelihoods and long-term demand.

Key term	Meaning	Context
forest	ecosystem dominated by trees	Tropical forests often support high biodiversity.
deforestation	conversion of forest to nonforest land use	Community forestry can link local benefits with stewardship.
degradation	decline in ecological quality without complete conversion	Community forestry can link local benefits with stewardship.
sustainable forestry	forest use maintaining long-term ecological and productive	Community forestry can link local benefits with stewardship.

REAL-WORLD EXAMPLES

- Tropical forests often support high biodiversity.
- Community forestry can link local benefits with stewardship.

FIELD / DATA TASK

Distinguish deforestation from forest degradation and give one example of each.

State the variable or evidence you would collect.

Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Forests and Forest Resources.

Define two key terms from this page.

Give one environmental benefit or risk linked to this topic.

Apply the concept to a local or Madagascar example.

GUIDING IDEA

Food systems connect soil, water, crops, livestock, energy, labor, transport, markets and waste.

CORE CONCEPTS

- Intensive agriculture can raise yields but may increase fertilizer, pesticide, water and energy use.
- Agroecological methods use ecological principles such as diversity, soil cover and nutrient cycling.
- Food loss and waste increase environmental pressure without improving nutrition.
- Sustainable food systems consider production, distribution, access and resilience.

Key term	Meaning	Context
food system	network linking food production, distribution, consumption and	Legume rotations can add biologically fixed nitrogen.
agroecology	application of ecological principles to agriculture	Mulching can reduce evaporation and erosion.
crop rotation	planned sequence of crops over time	Mulching can reduce evaporation and erosion.
food waste	edible food discarded or lost	Mulching can reduce evaporation and erosion.

REAL-WORLD EXAMPLES

- Legume rotations can add biologically fixed nitrogen.
- Mulching can reduce evaporation and erosion.

FIELD / DATA TASK

Compare a monoculture system and a diversified farm using four environmental criteria.

State the variable or evidence you would collect.

Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Agriculture and Food Systems.

Define two key terms from this page.

Give one environmental benefit or risk linked to this topic.

Apply the concept to a local or Madagascar example.

GUIDING IDEA

Fisheries depend on reproduction, habitat quality, food webs and harvest pressure.

CORE CONCEPTS

- If harvest exceeds population renewal for long periods, stocks can decline.
- Bycatch can affect non-target species.
- Nursery habitats such as mangroves can be essential to fish populations.
- Management tools include size limits, seasonal closures, quotas, gear rules and protected areas.

Key term	Meaning	Context
fishery	harvesting system for aquatic organisms	Seasonal closure can protect spawning fish.
stock	harvested population or group	Selective gear can reduce accidental capture of non-target
bycatch	unintended catch of non-target organisms	Selective gear can reduce accidental capture of non-target
quota	limit on allowable harvest	Selective gear can reduce accidental capture of non-target

REAL-WORLD EXAMPLES

- Seasonal closure can protect spawning fish.
- Selective gear can reduce accidental capture of non-target species.

FIELD / DATA TASK

Explain why protecting nursery habitat can be as important as regulating fishing effort.

State the variable or evidence you would collect.

Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Fisheries and Aquatic Resources.

Define two key terms from this page.

Give one environmental benefit or risk linked to this topic.

Apply the concept to a local or Madagascar example.

GUIDING IDEA

Mining supplies metals and minerals used in buildings, transport, electronics and energy systems, but can create environmental impacts.

CORE CONCEPTS

- Mining can disturb land, generate waste rock and alter water quality.
- Processing may require large amounts of water and energy.
- Environmental management includes site planning, tailings control, water treatment and rehabilitation.
- Recycling metals can reduce demand for some primary extraction.

Key term	Meaning	Context
ore	rock containing economically valuable minerals	Open-pit mines can transform landscapes.
tailings	waste material left after mineral processing	Mine rehabilitation may reshape land and re-establish
rehabilitation	making disturbed land stable and useful again	Mine rehabilitation may reshape land and re-establish
acid drainage	acidic water formed by oxidation of sulfide minerals	Mine rehabilitation may reshape land and re-establish

REAL-WORLD EXAMPLES

- Open-pit mines can transform landscapes.
- Mine rehabilitation may reshape land and re-establish vegetation.

FIELD / DATA TASK

List four stages of a mine life cycle and one environmental concern at each stage.

State the variable or evidence you would collect.

Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Minerals and Mining.

Define two key terms from this page.

Give one environmental benefit or risk linked to this topic.

Apply the concept to a local or Madagascar example.

GUIDING IDEA

Cities concentrate people, buildings, transport, energy use and waste, creating both environmental pressures and opportunities for efficient services.

CORE CONCEPTS

- Impervious surfaces increase runoff and can worsen flooding.
- Green infrastructure includes parks, street trees, wetlands, green roofs and permeable surfaces.
- Compact development can reduce travel distances but requires careful planning for housing and services.
- Urban environmental quality is closely linked to equity and public health.

Key term	Meaning	Context
urbanization	growth and expansion of urban settlements	Street trees can reduce heat and intercept rainfall.
impervious surface	surface that prevents water infiltration	Permeable paving allows more water to infiltrate.
green infrastructure	natural or designed green systems providing services	Permeable paving allows more water to infiltrate.
stormwater	rainwater flowing from built surfaces	Permeable paving allows more water to infiltrate.

REAL-WORLD EXAMPLES

- Street trees can reduce heat and intercept rainfall.
- Permeable paving allows more water to infiltrate.

FIELD / DATA TASK

Design three green-infrastructure features for a school or neighborhood.
State the variable or evidence you would collect.
Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Urban Land and Green Infrastructure.
Define two key terms from this page.
Give one environmental benefit or risk linked to this topic.
Apply the concept to a local or Madagascar example.

GUIDING IDEA

Tourism can support conservation and livelihoods but can also increase waste, water demand, habitat disturbance and carbon emissions.

CORE CONCEPTS

- Ecotourism aims to reduce harm, support conservation and benefit local communities.
- Carrying capacity in tourism considers ecological, social and infrastructure limits.
- Visitor management can use zoning, permits, trails and education.
- Benefits are more durable when local communities participate and receive fair returns.

Key term	Meaning	Context
ecotourism	nature-based tourism intended to support conservation and	Wildlife tourism can create incentives to conserve habitat.
carrying capacity	level of use a site can sustain without unacceptable impact	Heavy visitation can damage trails and disturb animals.
visitor management	tools controlling visitor behavior and distribution	Heavy visitation can damage trails and disturb animals.
community benefit	social or economic gain received locally	Heavy visitation can damage trails and disturb animals.

REAL-WORLD EXAMPLES

- Wildlife tourism can create incentives to conserve habitat.
- Heavy visitation can damage trails and disturb animals.

FIELD / DATA TASK

Write a five-rule environmental code for visitors to a sensitive natural site.

State the variable or evidence you would collect.

Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Tourism and Environmental Management.

Define two key terms from this page.

Give one environmental benefit or risk linked to this topic.

Apply the concept to a local or Madagascar example.

GUIDING IDEA

A circular economy aims to keep products and materials useful for longer and reduce waste and virgin resource extraction.

CORE CONCEPTS

- Reduce and redesign usually prevent more impact than end-of-pipe treatment.
- Reuse, repair, remanufacturing and recycling extend material life.
- Life-cycle thinking considers impacts from extraction through production, use and disposal.
- Circular systems still require energy and cannot eliminate all environmental impacts.

Key term	Meaning	Context
circular economy	system keeping materials in use and reducing waste	Repairing electronics can delay replacement and reduce
life cycle	stages from raw material through use and end-of-life	Deposit-return systems can increase container recovery.
reuse	using an item again without major reprocessing	Deposit-return systems can increase container recovery.
remanufacturing	restoring used products or parts to useful condition	Deposit-return systems can increase container recovery.

REAL-WORLD EXAMPLES

- Repairing electronics can delay replacement and reduce e-waste.
- Deposit-return systems can increase container recovery.

FIELD / DATA TASK

Map the life cycle of a smartphone and identify three opportunities to reduce impact.

State the variable or evidence you would collect.

Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Resource Efficiency and Circular Economy.

Define two key terms from this page.

Give one environmental benefit or risk linked to this topic.

Apply the concept to a local or Madagascar example.

GUIDING IDEA

Air pollution includes gases and particles that harm health, ecosystems, materials or climate.

CORE CONCEPTS

- Primary pollutants are emitted directly; secondary pollutants form through atmospheric reactions.
- Fine particles can penetrate deep into the respiratory system.
- Sources include vehicles, industry, energy use, fires and household fuels.
- Control combines cleaner technology, fuel choices, standards, monitoring and transport planning.

Key term	Meaning	Context
air pollutant	substance in air at harmful concentration	Vehicle exhaust contributes nitrogen oxides and particles in
particulate matter	small solid or liquid particles suspended in air	Smoke from open burning can reduce local air quality.
primary pollutant	pollutant emitted directly	Smoke from open burning can reduce local air quality.
secondary pollutant	pollutant formed in the atmosphere	Smoke from open burning can reduce local air quality.

REAL-WORLD EXAMPLES

- Vehicle exhaust contributes nitrogen oxides and particles in cities.
- Smoke from open burning can reduce local air quality.

FIELD / DATA TASK

Separate five pollutants into primary or secondary categories and justify your choices.

State the variable or evidence you would collect.

Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Air Pollution.

Define two key terms from this page.

Give one environmental benefit or risk linked to this topic.

Apply the concept to a local or Madagascar example.

GUIDING IDEA

Water pollution occurs when substances or energy alter water quality enough to harm organisms or human uses.

CORE CONCEPTS

- Point sources come from identifiable outlets; nonpoint sources are diffuse across landscapes.
- Organic waste can increase biological oxygen demand.
- Toxic substances can bioaccumulate in organisms and biomagnify through food webs.
- Prevention, treatment and watershed management are complementary strategies.

Key term	Meaning	Context
point source	single identifiable source of pollution	A wastewater pipe is a point source.
nonpoint source	diffuse pollution from many sources	Runoff from many farms is a nonpoint source.
BOD	oxygen used by microorganisms decomposing organic matter	Runoff from many farms is a nonpoint source.
bioaccumulation	buildup of substance within an organism over time	Runoff from many farms is a nonpoint source.

REAL-WORLD EXAMPLES

- A wastewater pipe is a point source.
- Runoff from many farms is a nonpoint source.

FIELD / DATA TASK

Explain the difference between bioaccumulation and biomagnification with a food-chain example.
State the variable or evidence you would collect.
Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Water Pollution.
Define two key terms from this page.
Give one environmental benefit or risk linked to this topic.
Apply the concept to a local or Madagascar example.

GUIDING IDEA

Eutrophication is nutrient enrichment of water that can trigger excessive plant or algal growth and oxygen depletion.

CORE CONCEPTS

- Nitrogen and phosphorus are common limiting nutrients in aquatic systems.
- Algal blooms can reduce light and alter food webs.
- When algae die, decomposition consumes dissolved oxygen.
- Reducing nutrient input is usually more effective than treating symptoms repeatedly.

Key term	Meaning	Context
nutrient enrichment	addition of nutrients beyond normal levels	Fertilizer runoff and untreated sewage can contribute nutrients.
algal bloom	rapid increase of algae	Fish kills may occur when dissolved oxygen becomes very low.
hypoxia	very low dissolved oxygen	Fish kills may occur when dissolved oxygen becomes very low.
phosphate	phosphorus-containing nutrient ion	Fish kills may occur when dissolved oxygen becomes very low.

REAL-WORLD EXAMPLES

- Fertilizer runoff and untreated sewage can contribute nutrients.
- Fish kills may occur when dissolved oxygen becomes very low.

FIELD / DATA TASK

Write the eutrophication process as a six-step cause-and-effect chain.
State the variable or evidence you would collect.
Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Eutrophication.
Define two key terms from this page.
Give one environmental benefit or risk linked to this topic.
Apply the concept to a local or Madagascar example.

GUIDING IDEA

Waste management aims to protect health and ecosystems while conserving materials and reducing disposal.

CORE CONCEPTS

- Waste prevention is generally preferred to reuse, recycling, recovery and disposal.
- Open dumping can contaminate soil and water and attract disease vectors.
- Sanitary landfills use engineering controls such as liners, leachate collection and gas management.
- Successful recycling depends on collection, sorting, markets and product design.

Key term	Meaning	Context
waste hierarchy	priority order from prevention to disposal	Separating organic waste can support composting.
landfill	engineered site for waste disposal	Landfill gas can contain methane from anaerobic decomposition.
leachate	contaminated liquid draining from waste	Landfill gas can contain methane from anaerobic decomposition.
recycling	processing discarded materials for new use	Landfill gas can contain methane from anaerobic decomposition.

REAL-WORLD EXAMPLES

- Separating organic waste can support composting.
- Landfill gas can contain methane from anaerobic decomposition.

FIELD / DATA TASK

Create a waste hierarchy for a school and give one action at each level.
State the variable or evidence you would collect.
Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Solid Waste Management.
Define two key terms from this page.
Give one environmental benefit or risk linked to this topic.
Apply the concept to a local or Madagascar example.

GUIDING IDEA

Plastics are useful, durable materials but poor management can lead to long-lived pollution on land and in water.

CORE CONCEPTS

- Single-use products can create large waste flows when collection systems are weak.
- Large plastic items can fragment into microplastics.
- Wildlife can be affected through entanglement or ingestion.
- Solutions include product redesign, reduction, reuse, collection, recycling and producer responsibility.

Key term	Meaning	Context
plastic pollution	accumulation of plastic in the environment	Plastic bags can block drainage systems.
microplastic	plastic particle generally smaller than 5 mm	Fishing gear can continue trapping wildlife after loss.
single-use	designed for one short use before disposal	Fishing gear can continue trapping wildlife after loss.
producer responsibility	policy making producers responsible for product end-of-life	Fishing gear can continue trapping wildlife after loss.

REAL-WORLD EXAMPLES

- Plastic bags can block drainage systems.
- Fishing gear can continue trapping wildlife after loss.

FIELD / DATA TASK

Identify which plastic problems require individual action, business action and government action.

State the variable or evidence you would collect.

Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Plastic Pollution.

Define two key terms from this page.

Give one environmental benefit or risk linked to this topic.

Apply the concept to a local or Madagascar example.

GUIDING IDEA

Hazardous wastes require special management because they may be toxic, corrosive, reactive, infectious or otherwise dangerous.

CORE CONCEPTS

- Mixing hazardous waste with ordinary waste can spread risk.
- Safe management includes identification, labeling, storage, transport, treatment and secure disposal.
- Medical waste and industrial chemicals may require specialized treatment.
- Pollution prevention can reduce hazardous waste at the source.

Key term	Meaning	Context
hazardous waste	waste with properties requiring special handling	Used solvents and some batteries contain hazardous
toxicity	capacity to cause harmful biological effects	Healthcare waste can include infectious materials and sharps.
corrosive	able to chemically damage materials or tissue	Healthcare waste can include infectious materials and sharps.
reactive	prone to dangerous chemical reaction	Healthcare waste can include infectious materials and sharps.

REAL-WORLD EXAMPLES

- Used solvents and some batteries contain hazardous substances.
- Healthcare waste can include infectious materials and sharps.

FIELD / DATA TASK

Design a simple segregation system for general, recyclable and hazardous waste in a laboratory.

State the variable or evidence you would collect.

Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Hazardous Waste.

Define two key terms from this page.

Give one environmental benefit or risk linked to this topic.

Apply the concept to a local or Madagascar example.

GUIDING IDEA

Electronic waste contains valuable materials as well as substances that can harm health and the environment when handled poorly.

CORE CONCEPTS

- Repair and reuse can extend product life.
- Informal burning or acid processing can release toxic substances.
- Formal recycling can recover metals under controlled conditions.
- Product design for disassembly can make repair and recycling easier.

Key term	Meaning	Context
e-waste	discarded electrical and electronic equipment	Old phones, computers and batteries are common e-waste
repair	restoring a product to working condition	Take-back schemes can channel devices into safer recycling
recovery	obtaining useful material or energy from waste	Take-back schemes can channel devices into safer recycling
design for disassembly	design enabling easy separation of parts	Take-back schemes can channel devices into safer recycling

REAL-WORLD EXAMPLES

- Old phones, computers and batteries are common e-waste items.
- Take-back schemes can channel devices into safer recycling systems.

FIELD / DATA TASK

Propose a school or community e-waste collection program with four practical steps.

State the variable or evidence you would collect.

Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Electronic Waste.

Define two key terms from this page.

Give one environmental benefit or risk linked to this topic.

Apply the concept to a local or Madagascar example.

GUIDING IDEA

Noise and artificial light can affect human health, wildlife behavior and ecosystem processes.

CORE CONCEPTS

- Chronic noise can interfere with communication and sleep.
- Artificial light at night can disrupt migration, feeding and reproduction in some species.
- Urban planning and technology can reduce unnecessary exposure.
- Pollution control is not limited to chemicals; energy and sensory disturbance also matter.

Key term	Meaning	Context
noise pollution	unwanted sound causing harm or disturbance	Road traffic can create persistent urban noise.
light pollution	excess or poorly directed artificial light	Bright coastal lighting can disorient some nocturnal animals.
exposure	people or assets located where hazard can affect them	Bright coastal lighting can disorient some nocturnal animals.
disturbance	disruption of normal activity or ecological process	Bright coastal lighting can disorient some nocturnal animals.

REAL-WORLD EXAMPLES

- Road traffic can create persistent urban noise.
- Bright coastal lighting can disorient some nocturnal animals.

FIELD / DATA TASK

Audit one area for noise or light pollution and propose two realistic improvements.

State the variable or evidence you would collect.

Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Noise and Light Pollution.

Define two key terms from this page.

Give one environmental benefit or risk linked to this topic.

Apply the concept to a local or Madagascar example.

GUIDING IDEA

Environmental monitoring measures conditions over time so that trends, hotspots and policy results can be evaluated.

CORE CONCEPTS

- Good monitoring uses clear indicators, consistent methods and quality control.
- Sampling design must represent the places and times relevant to the question.
- Baseline data provide a reference before a project or policy change.
- Data should be interpreted with uncertainty and natural variability in mind.

Key term	Meaning	Context
monitoring	repeated measurement to track condition or change	River monitoring may measure turbidity, nutrients and dissolved
indicator	measurable variable representing environmental condition	Air-quality stations record pollutant concentrations over time.
baseline	existing environmental condition before a project	Air-quality stations record pollutant concentrations over time.
sampling	selecting observations from a larger population or system	Air-quality stations record pollutant concentrations over time.

REAL-WORLD EXAMPLES

- River monitoring may measure turbidity, nutrients and dissolved oxygen at repeated sites.
- Air-quality stations record pollutant concentrations over time.

FIELD / DATA TASK

Design a monthly monitoring plan for one local environmental variable.
State the variable or evidence you would collect.
Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Pollution Monitoring.
Define two key terms from this page.
Give one environmental benefit or risk linked to this topic.
Apply the concept to a local or Madagascar example.

GUIDING IDEA

Environmental health studies how physical, chemical and biological exposures influence human well-being.

CORE CONCEPTS

- Risk depends on both hazard and exposure.
- Children, workers and other groups may have different vulnerability.
- Dose, duration and route of exposure affect outcomes.
- Preventive environmental measures can complement medical treatment by reducing exposure at the source.

Key term	Meaning	Context
hazard	potentially damaging event or process	Contaminated drinking water creates exposure through
exposure	people or assets located where hazard can affect them	Workers may need protective controls around hazardous dust or
risk	likelihood and severity of harm under exposure	Workers may need protective controls around hazardous dust or
dose	amount of a substance received	Workers may need protective controls around hazardous dust or

REAL-WORLD EXAMPLES

- Contaminated drinking water creates exposure through ingestion.
- Workers may need protective controls around hazardous dust or chemicals.

FIELD / DATA TASK

Explain why a dangerous substance may present low risk if exposure is effectively prevented.

State the variable or evidence you would collect.

Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Environmental Health.

Define two key terms from this page.

Give one environmental benefit or risk linked to this topic.

Apply the concept to a local or Madagascar example.

GUIDING IDEA

Weather describes short-term atmospheric conditions, while climate describes long-term patterns and variability.

CORE CONCEPTS

- Climate is influenced by latitude, elevation, oceans, land cover and atmospheric circulation.
- The climate system includes atmosphere, oceans, ice, land and living organisms.
- Natural variability occurs on many timescales.
- Long-term trends are evaluated using multiple records rather than single weather events.

Key term	Meaning	Context
weather	short-term atmospheric conditions	A single storm is weather; changing rainfall patterns over
climate	long-term pattern and variability of weather	Coastal climates often differ from inland climates at the same
variability	natural or other fluctuation around average conditions	Coastal climates often differ from inland climates at the same
climate system	interacting atmosphere, ocean, ice, land and biosphere	Coastal climates often differ from inland climates at the same

REAL-WORLD EXAMPLES

- A single storm is weather; changing rainfall patterns over decades relate to climate.
- Coastal climates often differ from inland climates at the same latitude.

FIELD / DATA TASK

Write three differences between weather and climate and give one example of each.

State the variable or evidence you would collect.

Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Weather, Climate and Climate Systems.

Define two key terms from this page.

Give one environmental benefit or risk linked to this topic.

Apply the concept to a local or Madagascar example.

GUIDING IDEA

Greenhouse gases absorb and re-emit infrared radiation, keeping Earth warmer than it would otherwise be.

CORE CONCEPTS

- The natural greenhouse effect is essential for life as we know it.
- Human activities increase concentrations of greenhouse gases such as carbon dioxide, methane and nitrous oxide.
- The enhanced greenhouse effect changes Earth's energy balance.
- Different gases vary in lifetime, concentration and warming influence.

Key term	Meaning	Context
greenhouse effect	warming caused by gases absorbing outgoing infrared	Fossil-fuel combustion increases carbon dioxide.
infrared radiation	longwave energy emitted by Earth	Agriculture and waste can contribute methane emissions.
carbon dioxide	major greenhouse gas from carbon cycling and combustion	Agriculture and waste can contribute methane emissions.
methane	potent greenhouse gas from biological and fossil sources	Agriculture and waste can contribute methane emissions.

REAL-WORLD EXAMPLES

- Fossil-fuel combustion increases carbon dioxide.
- Agriculture and waste can contribute methane emissions.

FIELD / DATA TASK

Draw a simple diagram showing incoming solar radiation and outgoing infrared radiation.

State the variable or evidence you would collect.

Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of The Greenhouse Effect.

Define two key terms from this page.

Give one environmental benefit or risk linked to this topic.

Apply the concept to a local or Madagascar example.

GUIDING IDEA

Climate change is assessed using temperature records, ocean measurements, ice observations, sea level, ecological responses and other indicators.

CORE CONCEPTS

- No single indicator tells the entire story.
- Independent datasets strengthen conclusions when they show consistent patterns.
- Attribution studies compare observed changes with modeled natural and human influences.
- Regional impacts can differ even within a global warming trend.

Key term	Meaning	Context
climate indicator	measurement revealing climate state or change	Glacier retreat is one indicator among many.
attribution	analysis of causes contributing to observed change	Changes in seasonal timing can affect flowering or migration.
sea level	average height of the ocean surface	Changes in seasonal timing can affect flowering or migration.
ocean warming	increase in ocean heat content	Changes in seasonal timing can affect flowering or migration.

REAL-WORLD EXAMPLES

- Glacier retreat is one indicator among many.
- Changes in seasonal timing can affect flowering or migration.

FIELD / DATA TASK

Why is a combination of independent indicators stronger than one dataset alone?

State the variable or evidence you would collect.

Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Climate Change Evidence.

Define two key terms from this page.

Give one environmental benefit or risk linked to this topic.

Apply the concept to a local or Madagascar example.

GUIDING IDEA

Climate change can alter rainfall, evaporation, drought risk, floods, crop suitability and water demand.

CORE CONCEPTS

- Warmer air can increase evaporation and change soil moisture.
- Extreme rainfall can increase flood and erosion risk.
- Agricultural impacts depend on crop type, soil, water access and adaptation.
- Water and food systems are linked, so one shock can affect multiple sectors.

Key term	Meaning	Context
drought	prolonged period of unusually low water availability	A prolonged drought can reduce reservoir storage and crop
flood	overflow of water onto normally dry land	Intense rain after dry periods can cause erosion and flash
soil moisture	water held in soil	Intense rain after dry periods can cause erosion and flash
food security	reliable access to sufficient safe nutritious food	Intense rain after dry periods can cause erosion and flash

REAL-WORLD EXAMPLES

- A prolonged drought can reduce reservoir storage and crop yields.
- Intense rain after dry periods can cause erosion and flash flooding.

FIELD / DATA TASK

Create a cause-effect map connecting climate, water, agriculture and food prices.

State the variable or evidence you would collect.

Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Climate Impacts on Water and Food.

Define two key terms from this page.

Give one environmental benefit or risk linked to this topic.

Apply the concept to a local or Madagascar example.

GUIDING IDEA

Species respond to climate change through shifts in range, timing, behavior, survival and ecological interactions.

CORE CONCEPTS

- Some species can move to suitable climates, while others are blocked by geography or habitat fragmentation.
- Seasonal timing can become mismatched between interacting species.
- Coral reefs are sensitive to heat stress and ocean changes.
- Reducing non-climate pressures can improve ecosystem resilience.

Key term	Meaning	Context
range shift	change in geographic distribution of a species	Mountain species may have limited space to move upward.
phenology	timing of seasonal biological events	Earlier flowering can affect pollinator interactions.
resilience	capacity to absorb disturbance and recover	Earlier flowering can affect pollinator interactions.
heat stress	harm caused by excessive temperature	Earlier flowering can affect pollinator interactions.

REAL-WORLD EXAMPLES

- Mountain species may have limited space to move upward.
- Earlier flowering can affect pollinator interactions.

FIELD / DATA TASK

Explain how habitat corridors can support climate adaptation for wildlife.
State the variable or evidence you would collect.
Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Climate Impacts on Biodiversity.
Define two key terms from this page.
Give one environmental benefit or risk linked to this topic.
Apply the concept to a local or Madagascar example.

GUIDING IDEA

Mitigation reduces greenhouse-gas emissions or increases removal of carbon from the atmosphere.

CORE CONCEPTS

- Energy efficiency, renewable energy, transport changes and reduced deforestation can lower emissions.
- Protecting and restoring ecosystems can maintain or increase carbon storage.
- Mitigation options differ in cost, speed, co-benefits and limitations.
- Deep reductions usually require action across multiple sectors rather than one technology.

Key term	Meaning	Context
mitigation	action reducing greenhouse-gas emissions or increasing	Solar power can reduce fossil-fuel electricity where conditions
decarbonization	reducing carbon intensity of energy and economic activity	Forest protection can avoid emissions from land-use change.
carbon sink	reservoir absorbing more carbon than it releases	Forest protection can avoid emissions from land-use change.
energy efficiency	same service with less energy input	Forest protection can avoid emissions from land-use change.

REAL-WORLD EXAMPLES

- Solar power can reduce fossil-fuel electricity where conditions are suitable.
- Forest protection can avoid emissions from land-use change.

FIELD / DATA TASK

Rank five mitigation actions by likely speed of implementation in your community and explain your reasoning.

State the variable or evidence you would collect.

Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Mitigation of Climate Change.

Define two key terms from this page.

Give one environmental benefit or risk linked to this topic.

Apply the concept to a local or Madagascar example.

GUIDING IDEA

Adaptation reduces vulnerability or takes advantage of changing conditions.

CORE CONCEPTS

- Adaptation can be structural, ecological, technological, behavioral or institutional.
- Good adaptation is based on local risks and vulnerable groups.
- Some measures create co-benefits, such as urban trees reducing both heat and runoff.
- Maladaptation occurs when a response increases risk elsewhere or in the future.

Key term	Meaning	Context
adaptation	adjustment reducing harm from climate impacts	Drought-resistant crops can reduce agricultural vulnerability.
vulnerability	susceptibility to climate harm	Mangrove restoration can support coastal protection and
resilience	capacity to absorb disturbance and recover	Mangrove restoration can support coastal protection and
maladaptation	response that increases vulnerability or emissions	Mangrove restoration can support coastal protection and

REAL-WORLD EXAMPLES

- Drought-resistant crops can reduce agricultural vulnerability.
- Mangrove restoration can support coastal protection and biodiversity.

FIELD / DATA TASK

Design three adaptation measures for a coastal town exposed to storms and sea-level rise.

State the variable or evidence you would collect.

Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Adaptation to Climate Change.

Define two key terms from this page.

Give one environmental benefit or risk linked to this topic.

Apply the concept to a local or Madagascar example.

GUIDING IDEA

Climate justice examines how causes, impacts, responsibilities and response costs are distributed among people, places and generations.

CORE CONCEPTS

- Communities with low emissions can still face high climate vulnerability.
- Historical emissions and present capacity are often considered in responsibility debates.
- Adaptation finance raises questions of fairness and access.
- Justice also includes future generations who cannot participate in current decisions.

Key term	Meaning	Context
climate justice	fairness in causes, impacts and responses to climate change	Small island communities may face severe coastal risk despite
equity	fairness considering different needs and circumstances	Low-income households may have fewer resources to recover
responsibility	obligation linked to contribution or capacity	Low-income households may have fewer resources to recover
vulnerability	susceptibility to climate harm	Low-income households may have fewer resources to recover

REAL-WORLD EXAMPLES

- Small island communities may face severe coastal risk despite small contributions to global emissions.
- Low-income households may have fewer resources to recover after disasters.

FIELD / DATA TASK

Give two reasons why climate policy is not only a scientific issue but also an ethical and social issue.

State the variable or evidence you would collect.

Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Climate Justice.

Define two key terms from this page.

Give one environmental benefit or risk linked to this topic.

Apply the concept to a local or Madagascar example.

GUIDING IDEA

Disaster risk depends on hazards, exposure, vulnerability and capacity, not on hazards alone.

CORE CONCEPTS

- Land-use planning can reduce exposure to floods, landslides and coastal hazards.
- Early warning systems can reduce loss of life.
- Healthy ecosystems such as wetlands can sometimes reduce hazard impacts.
- Preparedness and recovery planning are part of resilience.

Key term	Meaning	Context
hazard	potentially damaging event or process	Building on a floodplain increases exposure.
exposure	people or assets located where hazard can affect them	Mangroves can reduce some coastal wave energy.
vulnerability	susceptibility to climate harm	Mangroves can reduce some coastal wave energy.
capacity	maximum rated output under specified conditions	Mangroves can reduce some coastal wave energy.

REAL-WORLD EXAMPLES

- Building on a floodplain increases exposure.
- Mangroves can reduce some coastal wave energy.

FIELD / DATA TASK

Use the risk framework to analyze one cyclone, flood, drought or landslide scenario.

State the variable or evidence you would collect.

Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Disaster Risk Reduction.

Define two key terms from this page.

Give one environmental benefit or risk linked to this topic.

Apply the concept to a local or Madagascar example.

GUIDING IDEA

Stratospheric ozone absorbs much harmful ultraviolet radiation, while some pollutants can destroy ozone molecules.

CORE CONCEPTS

- Ozone high in the stratosphere is beneficial, while ground-level ozone is a harmful air pollutant.
- International cooperation reduced the use of many ozone-depleting substances.
- Atmospheric problems can cross borders, requiring coordinated policy.
- The ozone case is a useful example of science-policy interaction.

Key term	Meaning	Context
stratospheric ozone	ozone layer high in atmosphere absorbing UV	Refrigerant chemicals historically contributed to ozone depletion.
UV radiation	ultraviolet radiation from the Sun	Sun protection remains important even where ozone recovery is
ozone depletion	reduction of stratospheric ozone	Sun protection remains important even where ozone recovery is
ground-level ozone	secondary air pollutant near Earth surface	Sun protection remains important even where ozone recovery is

REAL-WORLD EXAMPLES

- Refrigerant chemicals historically contributed to ozone depletion.
- Sun protection remains important even where ozone recovery is occurring.

FIELD / DATA TASK

Explain why stratospheric ozone and ground-level ozone have very different environmental meanings.

State the variable or evidence you would collect.

Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Ozone Layer and Atmospheric Protection.

Define two key terms from this page.

Give one environmental benefit or risk linked to this topic.

Apply the concept to a local or Madagascar example.

GUIDING IDEA

Energy systems convert primary resources into useful services such as lighting, transport, heating and industrial power.

CORE CONCEPTS

- Fossil fuels are energy-dense but release greenhouse gases and air pollutants when burned.
- Renewable sources include solar, wind, hydropower, geothermal and sustainably managed biomass.
- No energy source is impact-free; land, materials, water, reliability and cost all matter.
- Energy planning compares full life-cycle impacts rather than only operation.

Key term	Meaning	Context
primary energy	energy available in natural resources before conversion	Solar panels produce electricity without combustion during
fossil fuel	coal, oil or gas formed from ancient organic matter	Hydropower can provide low-carbon electricity but alter river
renewable energy	energy from naturally replenished flows	Hydropower can provide low-carbon electricity but alter river
life cycle	stages from raw material through use and end-of-life	Hydropower can provide low-carbon electricity but alter river

REAL-WORLD EXAMPLES

- Solar panels produce electricity without combustion during operation.
- Hydropower can provide low-carbon electricity but alter river ecosystems.

FIELD / DATA TASK

Create a comparison table for solar, wind, hydro and fossil-fuel electricity.
State the variable or evidence you would collect.
Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Energy Resources.
Define two key terms from this page.
Give one environmental benefit or risk linked to this topic.
Apply the concept to a local or Madagascar example.

GUIDING IDEA

Solar energy can be used directly for heat or converted to electricity with photovoltaic systems.

CORE CONCEPTS

- Solar output varies with daylight, weather, orientation and shading.
- Battery storage can shift electricity supply to later periods.
- Lifecycle impacts include mining, manufacturing, transport and end-of-life management.
- Distributed solar can serve remote areas where grid access is limited.

Key term	Meaning	Context
photovoltaic	important environmental science term	Rooftop solar can supply daytime electricity to buildings.
solar radiation	energy arriving from the Sun	Solar water heating can reduce demand for other fuels.
battery storage	electrochemical storage of electricity	Solar water heating can reduce demand for other fuels.
distributed generation	small power sources located near users	Solar water heating can reduce demand for other fuels.

REAL-WORLD EXAMPLES

- Rooftop solar can supply daytime electricity to buildings.
- Solar water heating can reduce demand for other fuels.

FIELD / DATA TASK

List five factors you would assess before installing solar power at a school.

State the variable or evidence you would collect.

Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Solar Energy.

Define two key terms from this page.

Give one environmental benefit or risk linked to this topic.

Apply the concept to a local or Madagascar example.

GUIDING IDEA

Wind turbines use moving air, while hydropower uses flowing or falling water to generate electricity.

CORE CONCEPTS

- Wind resources vary strongly by location and height.
- Large dams can change river flow, sediment movement and fish migration.
- Run-of-river hydropower usually stores less water than large reservoirs but still affects rivers.
- Site selection and environmental assessment are critical for both technologies.

Key term	Meaning	Context
wind turbine	machine converting wind energy to electricity	Coastal or ridge locations may have strong wind resources.
hydropower	electricity from moving or falling water	A dam can trap sediment that would otherwise move
reservoir	stored body of water behind a barrier	A dam can trap sediment that would otherwise move
run-of-river	hydropower with limited water storage	A dam can trap sediment that would otherwise move

REAL-WORLD EXAMPLES

- Coastal or ridge locations may have strong wind resources.
- A dam can trap sediment that would otherwise move downstream.

FIELD / DATA TASK

Compare two environmental advantages and two disadvantages of large hydropower.

State the variable or evidence you would collect.

Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Wind and Hydropower.

Define two key terms from this page.

Give one environmental benefit or risk linked to this topic.

Apply the concept to a local or Madagascar example.

GUIDING IDEA

Biomass stores solar energy in plant material and can be used for heat, electricity or fuels.

CORE CONCEPTS

- Bioenergy can be renewable if biomass regrows and harvesting is sustainable.
- Traditional fuelwood use can contribute to indoor air pollution and forest pressure.
- Land competition can occur when energy crops displace food production or natural habitats.
- Efficient stoves and sustainable fuel supply can reduce some impacts.

Key term	Meaning	Context
biomass	organic material used as resource or fuel	Crop residues can be used as an energy source in some
biofuel	fuel made from recently living material	Unsustainable charcoal production can contribute to forest
fuelwood	wood burned for energy	Unsustainable charcoal production can contribute to forest
sustainable harvest	use that does not exceed long-term regeneration	Unsustainable charcoal production can contribute to forest

REAL-WORLD EXAMPLES

- Crop residues can be used as an energy source in some systems.
- Unsustainable charcoal production can contribute to forest degradation.

FIELD / DATA TASK

Explain why biomass can be renewable in one situation but unsustainable in another.

State the variable or evidence you would collect.

Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Biomass and Bioenergy.

Define two key terms from this page.

Give one environmental benefit or risk linked to this topic.

Apply the concept to a local or Madagascar example.

GUIDING IDEA

Energy efficiency provides the same or improved service using less energy.

CORE CONCEPTS

- Efficient buildings reduce heating, cooling and lighting demand.
- Efficient motors, appliances and vehicles lower operating energy use.
- Efficiency often saves money over time but may require higher upfront investment.
- Behavior and system design matter alongside efficient equipment.

Key term	Meaning	Context
energy efficiency	same service with less energy input	LED lighting uses less electricity than older incandescent lamps
energy intensity	energy used per unit of activity or output	Building shade can reduce cooling demand in hot climates.
insulation	material reducing heat transfer	Building shade can reduce cooling demand in hot climates.
efficiency standard	minimum required energy performance	Building shade can reduce cooling demand in hot climates.

REAL-WORLD EXAMPLES

- LED lighting uses less electricity than older incandescent lamps for similar light output.
- Building shade can reduce cooling demand in hot climates.

FIELD / DATA TASK

Audit five energy uses in a classroom and identify one efficiency improvement for each.

State the variable or evidence you would collect.

Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Energy Efficiency.

Define two key terms from this page.

Give one environmental benefit or risk linked to this topic.

Apply the concept to a local or Madagascar example.

GUIDING IDEA

Sustainable development seeks to improve human well-being while maintaining ecological systems and opportunities for future generations.

CORE CONCEPTS

- Environmental, social and economic goals interact rather than existing as separate pillars.
- Development choices can create trade-offs and co-benefits.
- Long-term thinking helps prevent solutions that shift problems to future generations.
- Indicators should include well-being and environmental quality, not only economic output.

Key term	Meaning	Context
sustainable development	development meeting present needs while protecting future	Clean public transport can reduce pollution while improving
well-being	quality of life and human flourishing	Watershed protection can support both biodiversity and
trade-off	gain in one objective accompanied by loss in another	Watershed protection can support both biodiversity and
co-benefit	additional benefit from an action taken for another goal	Watershed protection can support both biodiversity and

REAL-WORLD EXAMPLES

- Clean public transport can reduce pollution while improving access.
- Watershed protection can support both biodiversity and drinking-water security.

FIELD / DATA TASK

Give one example of a project with environmental, social and economic benefits.

State the variable or evidence you would collect.

Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Sustainable Development.

Define two key terms from this page.

Give one environmental benefit or risk linked to this topic.

Apply the concept to a local or Madagascar example.

GUIDING IDEA

Sustainable urban systems aim to reduce resource use and pollution while improving housing, mobility, services and resilience.

CORE CONCEPTS

- Public transport, walking and cycling can reduce congestion and emissions.
- Compact mixed-use neighborhoods can shorten travel distances.
- Urban green space improves habitat, cooling and recreation.
- Waste, water and energy systems should be planned together.

Key term	Meaning	Context
sustainable city	urban area planned for low impact and high well-being	Bus rapid transit can move many people efficiently.
public transport	shared passenger transport system	Urban wetlands can store stormwater and provide habitat.
mixed use	development combining housing, work and services	Urban wetlands can store stormwater and provide habitat.
green space	vegetated public or private urban land	Urban wetlands can store stormwater and provide habitat.

REAL-WORLD EXAMPLES

- Bus rapid transit can move many people efficiently.
- Urban wetlands can store stormwater and provide habitat.

FIELD / DATA TASK

Sketch a sustainable neighborhood and label at least six environmental features.

State the variable or evidence you would collect.

Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Sustainable Cities.

Define two key terms from this page.

Give one environmental benefit or risk linked to this topic.

Apply the concept to a local or Madagascar example.

GUIDING IDEA

Consumption affects environmental pressure through the quantity, type and life cycle of goods and services used.

CORE CONCEPTS

- Needs, durability, repairability and sharing can matter as much as recycling.
- Environmental labels can help consumers, but claims should be credible and verifiable.
- Businesses influence impacts through design, sourcing and supply chains.
- Policy can shape choices through standards, information and infrastructure.

Key term	Meaning	Context
consumption	use of goods and services	Buying a durable repairable product can reduce replacement
durability	ability to last and remain functional	Shared transport can reduce the number of vehicles needed for
repairability	ease of restoring function	Shared transport can reduce the number of vehicles needed for
eco-label	label communicating verified environmental attributes	Shared transport can reduce the number of vehicles needed for

REAL-WORLD EXAMPLES

- Buying a durable repairable product can reduce replacement demand.
- Shared transport can reduce the number of vehicles needed for some users.

FIELD / DATA TASK

Choose one everyday product and identify environmental impacts at four life-cycle stages.

State the variable or evidence you would collect.

Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Sustainable Consumption.

Define two key terms from this page.

Give one environmental benefit or risk linked to this topic.

Apply the concept to a local or Madagascar example.

GUIDING IDEA

Environmental economics studies how incentives, prices, property rights and policies influence resource use and pollution.

CORE CONCEPTS

- Externalities occur when costs or benefits affect people not included in a market transaction.
- Pollution taxes or permit systems can create incentives to reduce emissions.
- Subsidies can support beneficial technologies but may also encourage harmful activities if poorly designed.
- Economic valuation can inform decisions but should not replace ethical and ecological judgment.

Key term	Meaning	Context
externality	cost or benefit affecting others outside a transaction	Air pollution from a factory can impose health costs on nearby
pollution tax	charge based on pollution released	A fee on waste disposal may encourage waste reduction if
subsidy	financial support altering costs or incentives	A fee on waste disposal may encourage waste reduction if
incentive	factor encouraging a behavior	A fee on waste disposal may encourage waste reduction if

REAL-WORLD EXAMPLES

- Air pollution from a factory can impose health costs on nearby residents.
- A fee on waste disposal may encourage waste reduction if alternatives exist.

FIELD / DATA TASK

Explain a negative externality using one local environmental example.
State the variable or evidence you would collect.
Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Environmental Economics.
Define two key terms from this page.
Give one environmental benefit or risk linked to this topic.
Apply the concept to a local or Madagascar example.

GUIDING IDEA

Footprint indicators estimate the resource use or emissions associated with people, products or activities.

CORE CONCEPTS

- A carbon footprint focuses on greenhouse-gas emissions.
- A water footprint estimates direct and indirect water use.
- Life-cycle assessment compares impacts across stages from raw materials to disposal.
- Footprints are useful for comparison but depend on system boundaries and assumptions.

Key term	Meaning	Context
carbon footprint	greenhouse-gas emissions associated with an activity or	Food transport is one part of a food product's carbon footprint,
water footprint	water use associated with an activity or product	Cotton clothing has water use in agriculture as well as
life-cycle assessment	analysis of impacts across product stages	Cotton clothing has water use in agriculture as well as
system boundary	limit defining what an analysis includes	Cotton clothing has water use in agriculture as well as

REAL-WORLD EXAMPLES

- Food transport is one part of a food product's carbon footprint, but production can also be important.
- Cotton clothing has water use in agriculture as well as manufacturing.

FIELD / DATA TASK

Define a system boundary for comparing reusable and single-use drinking bottles.

State the variable or evidence you would collect.

Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Environmental Footprints.

Define two key terms from this page.

Give one environmental benefit or risk linked to this topic.

Apply the concept to a local or Madagascar example.

GUIDING IDEA

Environmental impact assessment evaluates likely effects of a proposed project before major decisions are finalized.

CORE CONCEPTS

- Typical stages include screening, scoping, baseline study, impact prediction, mitigation, consultation and monitoring.
- Alternatives should be compared, including location, design and sometimes a no-project option.
- Mitigation follows a hierarchy: avoid, minimize, restore and compensate where appropriate.
- Public participation can reveal local knowledge and social concerns.

Key term	Meaning	Context
EIA	process assessing environmental effects before project	A road project may affect forests, water, noise and communities.
screening	deciding whether detailed assessment is required	A mine proposal may require water, biodiversity and
scoping	identifying important issues for assessment	A mine proposal may require water, biodiversity and
baseline	existing environmental condition before a project	A mine proposal may require water, biodiversity and

REAL-WORLD EXAMPLES

- A road project may affect forests, water, noise and communities.
- A mine proposal may require water, biodiversity and rehabilitation plans.

FIELD / DATA TASK

Draft a five-point scope for an EIA of a new road crossing a rural landscape.

State the variable or evidence you would collect.

Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Environmental Impact Assessment.

Define two key terms from this page.

Give one environmental benefit or risk linked to this topic.

Apply the concept to a local or Madagascar example.

GUIDING IDEA

Governments use laws, standards, taxes, permits, information, planning and voluntary agreements to influence environmental behavior.

CORE CONCEPTS

- Command-and-control rules set limits or required practices.
- Market-based tools change incentives through prices or tradable rights.
- Information tools improve transparency and consumer choice.
- Effective policy depends on monitoring, enforcement, institutional capacity and public trust.

Key term	Meaning	Context
regulation	binding rule set by authority	Vehicle emission standards can limit pollutants.
standard	required performance or quality level	Deposit-refund systems can encourage container return.
permit	official authorization subject to conditions	Deposit-refund systems can encourage container return.
market instrument	policy changing incentives through prices or tradable rights	Deposit-refund systems can encourage container return.

REAL-WORLD EXAMPLES

- Vehicle emission standards can limit pollutants.
- Deposit-refund systems can encourage container return.

FIELD / DATA TASK

Choose one pollution problem and compare a regulation with an economic incentive for controlling it.

State the variable or evidence you would collect.

Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Environmental Policy Instruments.

Define two key terms from this page.

Give one environmental benefit or risk linked to this topic.

Apply the concept to a local or Madagascar example.

GUIDING IDEA

Environmental research uses questions, hypotheses, variables, sampling and analysis to build reliable knowledge.

CORE CONCEPTS

- Observational studies describe patterns, while experiments test controlled changes where feasible.
- Replication improves confidence by showing whether a result is consistent.
- Sampling should reduce bias and represent the target system.
- Correlation alone does not prove causation.

Key term	Meaning	Context
hypothesis	testable proposed explanation or prediction	Quadrats can estimate plant abundance.
variable	factor measured or changed in a study	Repeated water samples can reveal seasonal changes.
sampling	selecting observations from a larger population or system	Repeated water samples can reveal seasonal changes.
replication	repeating observations or experiments	Repeated water samples can reveal seasonal changes.

REAL-WORLD EXAMPLES

- Quadrats can estimate plant abundance.
- Repeated water samples can reveal seasonal changes.

FIELD / DATA TASK

Write a testable hypothesis about vegetation cover and soil erosion.
State the variable or evidence you would collect.
Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Environmental Research Methods.
Define two key terms from this page.
Give one environmental benefit or risk linked to this topic.
Apply the concept to a local or Madagascar example.

GUIDING IDEA

Environmental decisions often rely on graphs, maps, tables and time series that must be interpreted carefully.

CORE CONCEPTS

- Always read axes, units, scale, time period and data source.
- Averages can hide variation and extreme values.
- Trend and correlation should not be confused with certainty about cause.
- Uncertainty ranges communicate limits of measurement or prediction.

Key term	Meaning	Context
time series	measurements ordered through time	A rising line may show a trend but not explain its cause.
trend	persistent direction of change	A map using different class intervals can change visual
scale	magnitude or spatial/temporal level of analysis	A map using different class intervals can change visual
uncertainty	range of plausible values or limits to knowledge	A map using different class intervals can change visual

REAL-WORLD EXAMPLES

- A rising line may show a trend but not explain its cause.
- A map using different class intervals can change visual interpretation.

FIELD / DATA TASK

Create a checklist of six questions to ask before trusting a graph.
State the variable or evidence you would collect.
Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Reading Environmental Data.
Define two key terms from this page.
Give one environmental benefit or risk linked to this topic.
Apply the concept to a local or Madagascar example.

GUIDING IDEA

Madagascar provides important examples of endemism, forest change, agriculture, protected areas and community-based environmental management.

CORE CONCEPTS

- Environmental conditions vary strongly from humid eastern forests to dry western and southern ecosystems.
- Many species have restricted ranges and depend on particular habitats.
- Land-use decisions are shaped by livelihoods, markets, infrastructure, governance and ecological conditions.
- Effective solutions need local context rather than one universal model.

Key term	Meaning	Context
Madagascar	large island nation with exceptional biodiversity and varied	Lemurs and baobabs are well-known examples of Madagascar's
endemism	important environmental science term	Watershed restoration can link soil conservation, water quality
land use	human use and management of land	Watershed restoration can link soil conservation, water quality
livelihood	means of obtaining food, income and necessities	Watershed restoration can link soil conservation, water quality

REAL-WORLD EXAMPLES

- Lemurs and baobabs are well-known examples of Madagascar's distinctive biodiversity.
- Watershed restoration can link soil conservation, water quality and local livelihoods.

FIELD / DATA TASK

Choose one Madagascar ecosystem and explain its environmental value, threats and two realistic management actions.
State the variable or evidence you would collect.
Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Madagascar: Biodiversity and Land Use.
Define two key terms from this page.
Give one environmental benefit or risk linked to this topic.
Apply the concept to a local or Madagascar example.

GUIDING IDEA

Madagascar faces environmental risks linked to cyclones, floods, droughts, coastal change and livelihood vulnerability.

CORE CONCEPTS

- Risk varies by region, exposure, infrastructure and access to resources.
- Mangroves, dunes and wetlands can contribute to coastal protection while supporting biodiversity.
- Early warning, resilient infrastructure and land-use planning can reduce disaster impacts.
- Adaptation works best when scientific information and local experience are combined.

Key term	Meaning	Context
cyclone	rotating tropical storm system	Coastal communities may depend on fisheries while also facing
coastal resilience	ability of coastal systems and communities to withstand and	Restored vegetation can reduce erosion in some catchments.
mangrove	salt-tolerant coastal forest	Restored vegetation can reduce erosion in some catchments.
early warning	system providing timely hazard information	Restored vegetation can reduce erosion in some catchments.

REAL-WORLD EXAMPLES

- Coastal communities may depend on fisheries while also facing storm risk.
- Restored vegetation can reduce erosion in some catchments.

FIELD / DATA TASK

Prepare a resilience plan with actions before, during and after a cyclone.

State the variable or evidence you would collect.

Identify one source of uncertainty or limitation.

CHECKPOINT QUESTIONS

Explain the main idea of Madagascar: Climate, Coasts and Resilience.

Define two key terms from this page.

Give one environmental benefit or risk linked to this topic.

Apply the concept to a local or Madagascar example.

LINKING ENVIRONMENTAL SYSTEMS

- Environmental problems rarely have one cause: identify drivers, pressures, environmental changes, impacts and responses.
- Use arrows to connect atmosphere, water, soil, organisms and human activities.
- Look for feedback loops: a change may reinforce itself or trigger a stabilizing response.
- Separate stocks from flows when reasoning about resources or pollutants.
- Always state the spatial and time scale of the system you are analyzing.

SYNOPTIC PRACTICE

- Trace deforestation -> soil erosion -> river sediment -> coastal ecosystem impact.
- Trace urban growth -> impervious surfaces -> stormwater -> flood risk -> green infrastructure response.
- Trace fertilizer use -> runoff -> eutrophication -> hypoxia -> fisheries impact.
- Trace fossil-fuel use -> emissions -> climate change -> drought risk -> adaptation.

SYSTEMS VOCABULARY

Stock: quantity stored.

Flow: rate of movement.

Feedback: process affecting future change.

Threshold: point where system behavior changes.

Resilience: capacity to recover.

EXAM TASKS

Draw one cause-effect chain.

Add one feedback loop.

State one uncertainty.

Propose one intervention point.

COMPARE AND CONNECT

- Biodiversity can be measured at genetic, species and ecosystem levels.
- Major pressures include habitat loss, overuse, invasive species, pollution and climate change.
- Conservation methods include protected areas, corridors, restoration, sustainable use, ex situ conservation and community management.
- Resource sustainability depends on renewal rate, demand, technology, access and governance.
- Water, soil, forests, fisheries and agriculture are connected through landscapes and watersheds.

CASE-STUDY QUESTIONS

- Why does endemism increase conservation importance?
- How can forest restoration support both biodiversity and water quality?
- When can a renewable resource become depleted?
- Why is community participation important in conservation?
- How can agroecology reduce environmental pressure while maintaining production?

MADAGASCAR FOCUS

Endemic biodiversity.
Forest fragmentation.
Lemur conservation.
Watershed restoration.
Cyclone and coastal resilience.

COMMAND WORDS

Describe: give features.
Explain: show how/why.
Compare: similarities and differences.
Evaluate: judge using evidence.

POLLUTION CHAINS

- Identify source -> pathway -> exposure -> effect -> control.
- Distinguish point from nonpoint pollution and primary from secondary air pollutants.
- Use BOD and dissolved oxygen correctly when explaining organic water pollution.
- Explain bioaccumulation within organisms and biomagnification across trophic levels.
- Apply the waste hierarchy before choosing disposal.

CLIMATE CHAINS

- Distinguish weather from climate.
- Explain the greenhouse effect before discussing climate change.
- Separate mitigation from adaptation.
- Use hazard, exposure and vulnerability when explaining disaster risk.
- Consider climate justice when impacts and responsibilities are unevenly distributed.

PRACTICE

Explain eutrophication in six steps.
Compare mitigation and adaptation.
Give one co-benefit of urban trees.
Explain why climate evidence uses multiple indicators.

KEY TERMS

BOD
hypoxia
greenhouse effect
mitigation
adaptation
vulnerability

ENERGY COMPARISON

Option	Strength	Important limitation
Solar	Low operating emissions; variable output	Land/materials/storage considerations
Wind	Low operating emissions; scalable	Variable; site and wildlife considerations
Hydro	Dispatchable in many systems	River and sediment impacts
Biomass	Can use renewable biological material	Land, air pollution and harvest sustainability

SUSTAINABILITY PRINCIPLES

- Reduce demand before expanding supply where practical.
- Use life-cycle thinking rather than focusing on one stage.
- Prefer prevention, efficiency, durability, repair and circular material flows.
- Look for co-benefits across environment, health, livelihoods and resilience.
- State trade-offs honestly; no environmental option is impact-free.

PRACTICE

Compare two energy sources.
Define an externality.
Explain one circular-economy strategy.
Give one sustainable-city intervention.

DECISION CRITERIA

Environmental impact
Cost and affordability
Reliability
Equity
Local suitability
Long-term resilience

INVESTIGATION METHOD

- 1. State a clear research question.
- 2. Write a testable hypothesis where appropriate.
- 3. Define independent, dependent and controlled variables.
- 4. Choose a representative sampling strategy.
- 5. Record units and repeat measurements.
- 6. Present data using suitable tables, graphs or maps.
- 7. Interpret patterns and uncertainty.
- 8. Evaluate limitations and improvements.

GRAPH AND TABLE CHECKLIST

- Read title, axes, units, scale and time period.
- Identify overall trend and important exceptions.
- Do not infer causation from correlation alone.
- Compare relative and absolute change carefully.
- Check whether averages hide variation.
- State uncertainty when evidence is incomplete.

FIELD METHODS

Quadrat sampling
Transects
Water-quality testing
Soil observation
Species counts
Questionnaires

DATA WORDS

Mean
Median
Range
Trend
Correlation
Outlier
Uncertainty

SECTION A - SHORT ANSWERS

- Define ecosystem, biodiversity, carrying capacity, eutrophication and adaptation.
- Explain one difference between renewable and nonrenewable resources.
- Distinguish point and nonpoint water pollution.
- Explain why habitat fragmentation can reduce biodiversity.
- State two environmental benefits of mangroves.
- Explain the difference between mitigation and adaptation.

SECTION B - STRUCTURED QUESTIONS

- A lake receives fertilizer runoff. Explain the likely ecological sequence and propose two controls.
- A town experiences more flooding after rapid urban growth. Analyze likely causes and propose green-infrastructure solutions.
- Compare solar power and hydropower using climate, biodiversity, reliability and cost considerations.
- Design a monitoring plan for forest restoration.

SECTION C - ESSAYS

- "Biodiversity conservation succeeds only when local livelihoods are considered." Discuss.
- "The most sustainable waste is the waste that is never created." Evaluate using the waste hierarchy.
- "Climate change is a systems problem requiring both mitigation and adaptation." Discuss.

TIME STRATEGY

Define first.
Use cause-effect chains.
Give real examples.
Evaluate trade-offs.

ESSAY STRUCTURE

Introduction
Mechanism/evidence
Counterpoint or limitation
Case study
Reasoned conclusion

SHORT-ANSWER CORE POINTS

- Ecosystem: organisms plus physical environment and their interactions.
- Biodiversity: variety at genetic, species and ecosystem levels.
- Carrying capacity: population level supportable over time under given conditions.
- Eutrophication: nutrient enrichment leading to excessive growth and often oxygen depletion.
- Adaptation: adjustment reducing harm from climate impacts.
- Renewable resources replenish on useful timescales but can still be overused.

STRUCTURED QUESTION GUIDANCE

- Lake runoff: nutrients -> algal bloom -> decomposition -> oxygen decline; controls include nutrient management and wastewater treatment.
- Urban flooding: more impervious surfaces -> faster runoff; use permeable surfaces, wetlands, detention, trees and drainage planning.
- Energy comparison: use consistent criteria and state trade-offs rather than declaring one option universally best.
- Restoration monitoring: include baseline, repeated plots, survival, species diversity, soil condition and photo records.

ESSAY MARKING GUIDE

- Accurate definitions and mechanisms.
- Use of environmental evidence or case studies.
- Clear comparison of benefits, costs and limitations.
- Recognition of uncertainty and trade-offs.
- Direct conclusion answering the question.

SELF-CHECK

Did I answer the exact question?
Did I explain causes, not just list them?
Did I use a case study?
Did I evaluate?

IMPROVE

Add data where available.
Use correct units.
State scale and time period.
Separate evidence from opinion.

ESSENTIAL GLOSSARY

Ecosystem	living organisms interacting with their physical environment
Biodiversity	variety of genes, species and ecosystems
Resilience	capacity to absorb disturbance and recover
Sustainability	maintaining environmental and social capacity over time
Eutrophication	nutrient enrichment causing excessive biological growth
Mitigation	action reducing environmental harm or greenhouse-gas emissions
Adaptation	adjustment reducing vulnerability to impacts
Externality	cost or benefit affecting others outside a transaction
EIA	assessment of likely project impacts before major decisions
Life cycle	stages from resource extraction through use and end-of-life

FINAL CHECKLIST

- I can explain ecosystem structure, energy flow and nutrient cycles.
- I can analyze biodiversity threats and conservation responses.
- I can compare water, soil, forest, fisheries and mineral resource management.
- I can explain major pollution pathways and waste strategies.
- I can distinguish climate mitigation, adaptation and disaster risk reduction.
- I can compare energy options using life-cycle and sustainability criteria.
- I can design a simple environmental investigation and interpret data.
- I can apply environmental science to Madagascar and other real-world case studies.

END OF COURSEBOOK

Environmental science connects evidence, systems thinking and responsible decision-making.

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